Theory of Computation 1-7 Review of Mathematical Theory

Let us discuss each one by one -

1) Negation

horizontal bar on letter.

S	1s
T	F.
F	T

ii Conjunction (AND)

If A and B are two propositions then conjunction of A and B is denoted by A ^ B. The truth table for this connective is

A	B	A ^ B
T	T	T
T	F	F
F	T	F
F	F	F

iii) Disjunction OR)

If A and B are two propositions then disjunction of A and B is denoted by AvB. The truth table for this connective is

A	B	AvB
T	T	T
T	F	T
F	T	T
F	F	F

v implication

byABorA=B.

The truth table for this connective is

A	B	AB
T	T	T
T	F	F
F	T	T
F	F	T

For example: Find truth value for following proposition.

If an apple is not green then it must be red

LetA denotes An apple is not green and B denotes It must be redThe statement

A is false and B is true. Hence the proposition is true.

v)f and only if

If A and B are two propositions then A if and only if B. It can be denoted as AB

The truth table will be

A	8	AB
T	T	T
T	F	F
F	T	F
F	F	T

For exampleI will go to the school if and only if it is not raining.Here statement A is I will go to the school. The statement B is It is not raining Thus the resultant proposition is false.

1.5 Relations

Relaonship is a major aspect between two objects even this is true in our real life

form pair based on this certain relationship.

clements.

pair is chosen from a set called range.

Revlow of Mathomatical Theory

Properties of Relations

A relation R on set S is

1.Reflexive if iRi for all i in S.

2.Irreflexive if iRi is false for all i in S.

3.Transitive if iRj and jRk imply iRk

4.Symmetric if iRj implies jRi.

5.Asymmetric if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive

For example:ifA=ab then

reflexive relation R can be=a a)b,b)

irreflexive relation R can be =ab

transitive relation R can be=ab.b.aaa1

symmetric relation R can be=ab),b,a1

asymmetric relation R can be=a,b)

Equivalent Relation

A relation is said to be cquivalence relation if it is reflexive, symmetric and transitive

over some set S.

Suppose R is a set of relations and S is the set of elements.

For example : S is the set of lines in a plane and R is the relation of lines intersecting

to each other.

A relation is said to be antisymmetric if a,be R and b,ae R then a=a.

Example 1.5:Determine whether R is equivalence relation or not where

A=01,21R=/0010)11).2,2)2,1

Solution The R is reflexive because 0,01.1)2.2E R.

Where as R is not symmetric because (0, 1) is not in R whereas (1, 0) is in R.

Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given, it may not be reflexive or transitive

By adding some pairs we make the relation as reflexive or transitive.

For exampleLet ab).bc)aa)bb